

# Concept Plans and Spatial Organization Diagrams

How the park is organised — the crescent, the radial alleys, and the rooms between them.

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# Concept Development

## Al Safa 2 Park — Alternatives, Evaluation & Final Concept Selection

**AI-GENERATED DRAFT — FOR REVIEW.** Three distinct concept alternatives are developed and scored, each responding differently to Phase 3's objectives.

### 4.1 Research Inspiration

- **Local context:** Dubai's Shamal wind pattern, palm/Ghaf shade traditions, sikka (shaded alleyway) urban form.
- **International precedents (typology reference, not copied):** shaded desert park strategies using canopy structures and misting (Middle East precedent), linear park zoning with clear activity bands (High Line-style legibility), biophilic play design for inclusive playgrounds.
- **Landscape trends:** climate-responsive/native xeriscaping, multi-generational activity zoning, park-as-microclimate-infrastructure thinking.

### 4.2 Mood & Design Language Direction

Material and form language: warm sand/stone tones, filtered dappled shade (not solid roofed structures everywhere), local Ghaf/palm planting as the dominant canopy species, and a restrained geometric language (straight + arced paths) echoing the site's elongated rectangular geometry rather than fighting it.

### 4.4 Concept Alternatives

|                 | Concept A — "Shaded Spine"                              | Concept B — "Canopy Village"                                                     | Concept C — "Cool Loop"                                                        |
|-----------------|---------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Organizing idea | One continuous shaded central spine connects all zones  | Cluster of discrete shaded "rooms" (play, fitness, social, quiet) around a plaza | A single perimeter shaded loop (jogging/walking) with activities nested inside |
| Circulation     | Linear, highly legible, single primary path             | Radial from central plaza to each room                                           | Circular loop + inner cross-paths                                              |
| Shade strategy  | Continuous overhead shade structure along the spine     | Shade concentrated per-room (tree clusters + pergolas)                           | Shade concentrated on the loop itself                                          |
| Best fits       | Wayfinding clarity, day/night activation along one axis | Distinct age-group zoning, event flexibility                                     | Fitness/wellness emphasis, continuous exercise use                             |
| Risk            | Areas off the spine may still be under-shaded           | Central plaza could become a pinch-point / less flexible for events              | Interior zones could feel disconnected from the loop's comfort                 |

### 4.5 AI-Assisted Exploration

All three concepts were tested against Phase 1.06's shadow-length data (6m canopy shadow ranges from ~0.2m at summer noon to ~19m at winter evening) to sanity-check shade-structure spacing assumptions before scoring. This is documented further in Phase 9 (AI Workflow).

### 4.6 Evaluation Matrix

| Criterion (weight) | A: Shaded Spine | B: Canopy Village | C: Cool Loop |
|--------------------|-----------------|-------------------|--------------|
| Function (25%)     | 8               | 7                 | 8            |

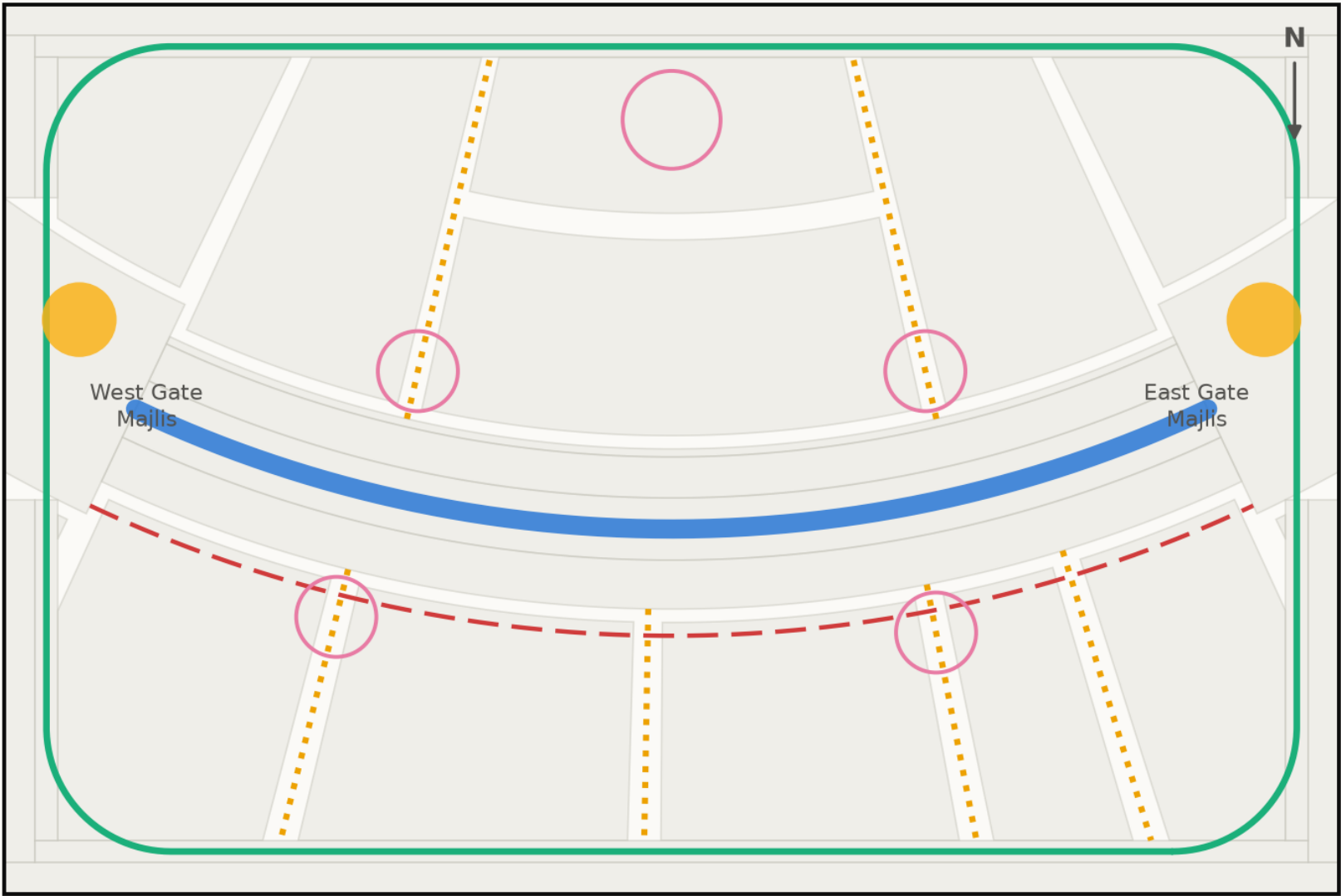
| Criterion (weight)               | A: Shaded Spine | B: Canopy Village | C: Cool Loop |
|----------------------------------|-----------------|-------------------|--------------|
| User Experience (25%)            | 8               | 9                 | 7            |
| Sustainability (20%)             | 7               | 8                 | 7            |
| Feasibility within AED 35M (20%) | 9               | 6                 | 8            |
| Innovation (10%)                 | 7               | 8                 | 6            |
| Weighted Total /10               | 7.85            | 7.55              | 7.35         |

### 4.7 Final Concept Selection

**Selected: Concept A — "Shaded Spine"**, with Concept B's room-based zoning logic merged in as secondary structure along the spine (best of both: legibility + distinct age-group zones). Justification: highest feasibility score within the fixed AED 35M budget, directly addresses Phase 2's #1 evidence-backed problem (summer thermal comfort) via one continuous engineered shade structure rather than dozens of smaller ones, and gives the AI Design Challenge submission the clearest single diagram to communicate (important per Evaluation Matrix criterion 6: Quality of Presentation, 5% weight, and criterion 4: Quality of Design and User Experience, 15%).

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



- Al Mamsha — primary route, shaded, step-free
- Al Sikkak — shaded alleys to the rooms
- Al Madar — 438 m running loop
- Service / emergency vehicle access
- Majlis — shaded rest, never more than 33 m from the walk

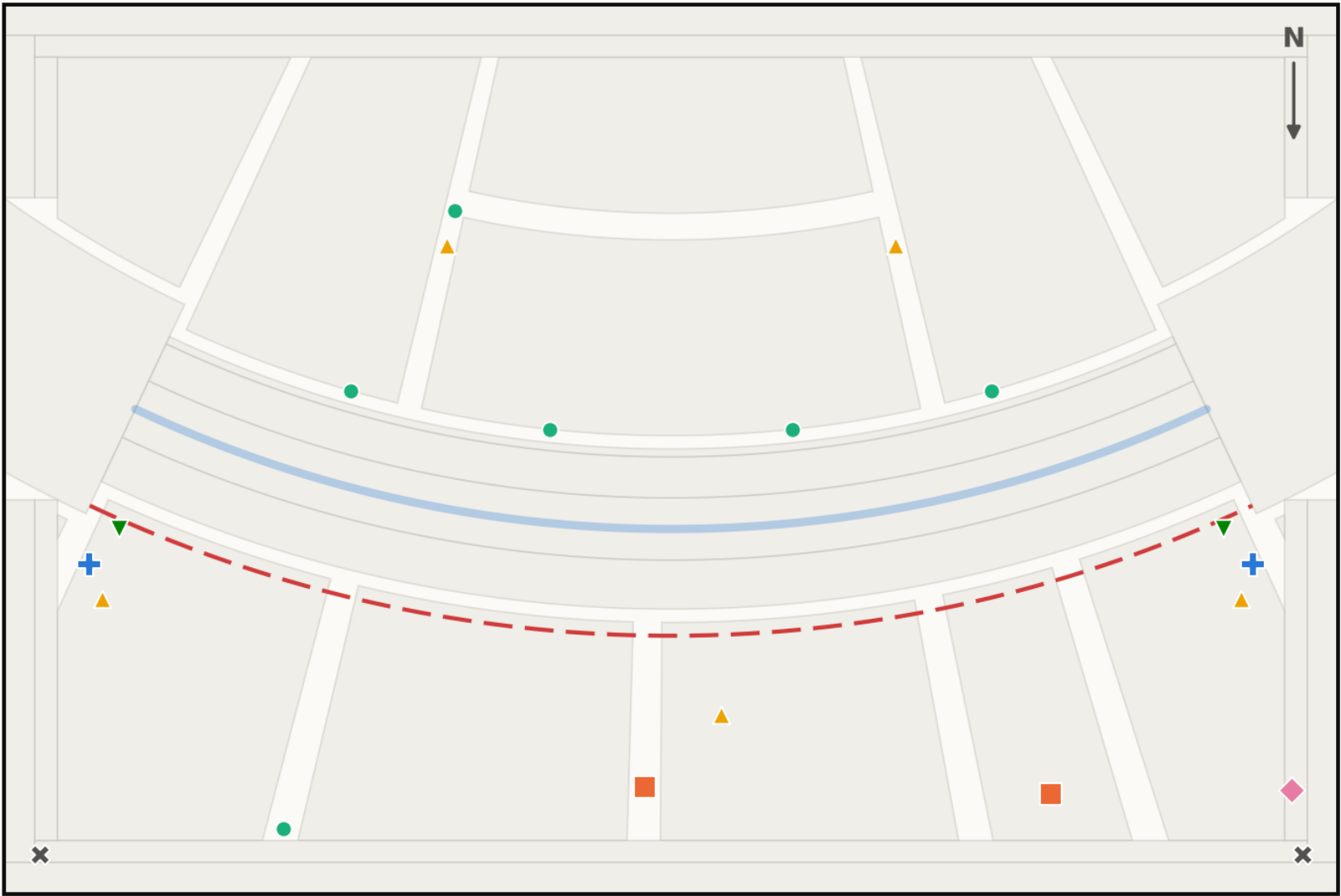
Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.  
Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Facilities

Technical drawing — to scale.

Commercial & service facilities

Where the park earns its running costs, and where it is serviced



20 m

- Service / emergency vehicle access

Commercial — souk kiosks and café (2)

Public restrooms (2)
- Drinking fountain (6)

Waste & recycling station (5)

Service & maintenance store (1)
- Bicycle parking (2)

Drop-off / pick-up bay (2)

20 facilities. Commercial floor sits on the convex south face (417 m2, 2.8% of the site), facing the plaza and the event lawn so that evening trade and programming reinforce each other — and away from the concave hollow, which is kept quiet. Every facility is reachable from the dashed service route without a vehicle entering the walk.

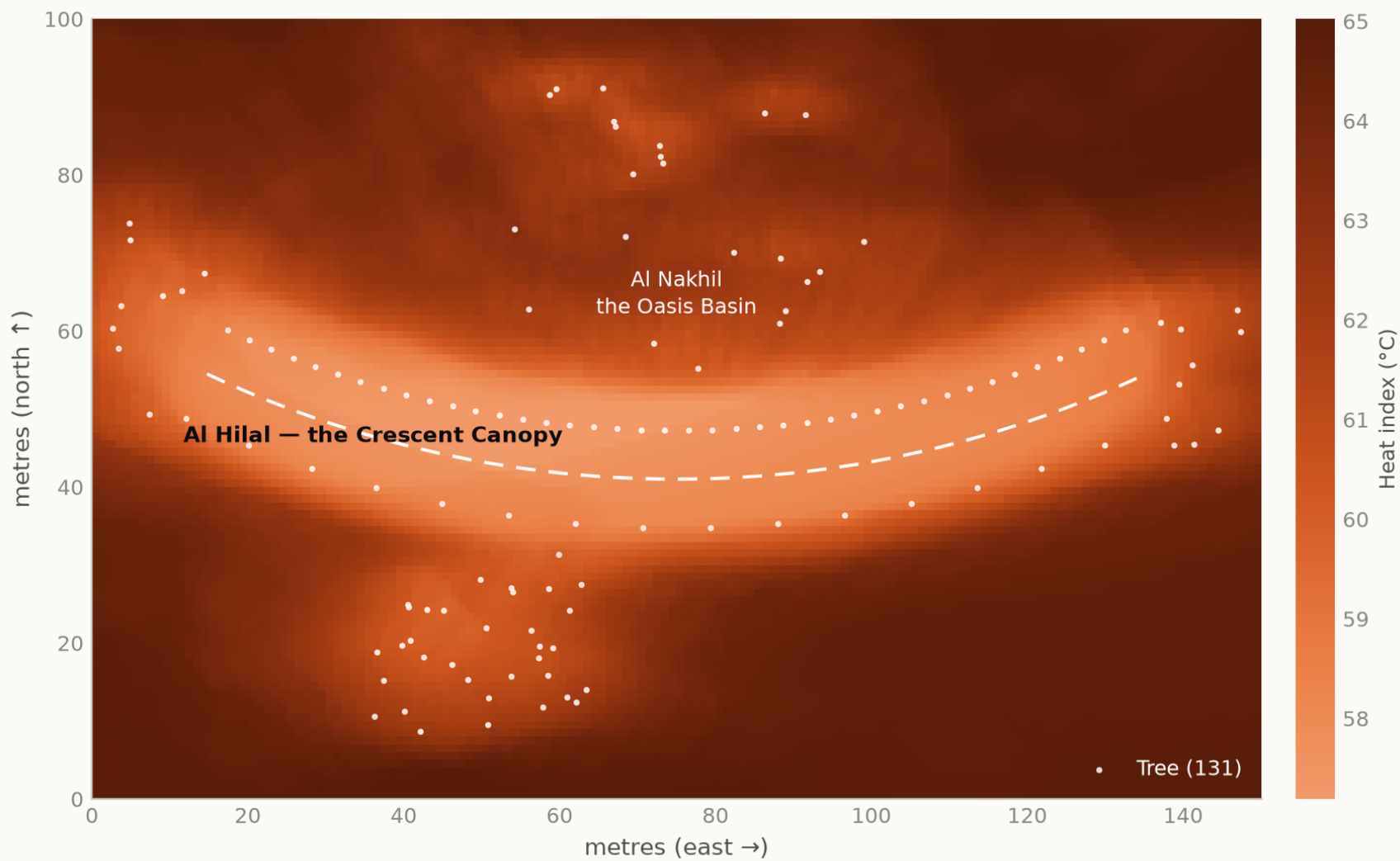
Source: src/plan.py — facility positions are arc coordinates, so they move with the crescent · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig04 site comfort map

Analysis output — computed from project data.

Predicted summer comfort across the site

July afternoon heat index per square metre — the shade strategy, tested



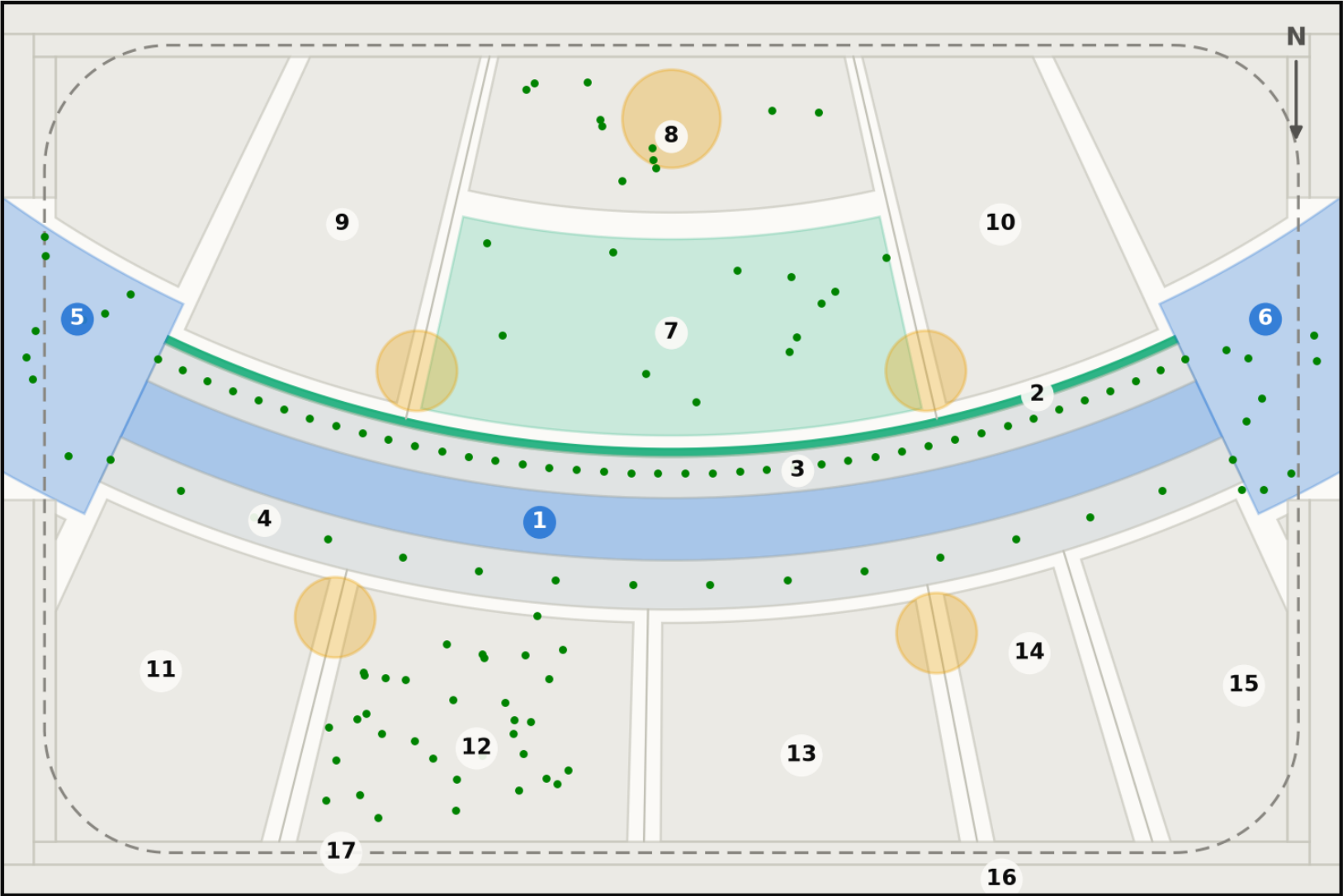
Lighter is cooler. The crescent reads as the coolest continuous route across the site, and its concave side is measurably cooler than its convex one — which is why the rooms people are asked to linger in are all placed there. White dots are the 131 trees.  
Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; NCM Dubai climate normals 1977-2015 (39-year record) · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

|                                      |                                     |          |
|--------------------------------------|-------------------------------------|----------|
| 1                                    | Al Mamsha — the Crescent Walk       | 871 m²   |
| 2                                    | Al Falaj — the water channel        | 105 m²   |
| 3                                    | Crescent Shade Margin (N)           | 549 m²   |
| 4                                    | Crescent Shade Margin (S)           | 715 m²   |
| 5                                    | West Gate Majlis                    | 439 m²   |
| 6                                    | East Gate Majlis                    | 439 m²   |
| 7                                    | Al Nakhil — the Oasis Basin         | 1,140 m² |
| 8                                    | Quiet Contemplation Garden          | 707 m²   |
| 9                                    | Children's Dune Play                | 1,267 m² |
| 10                                   | Family Picnic Grove                 | 1,267 m² |
| 11                                   | Outdoor Fitness Terrace             | 908 m²   |
| 12                                   | Native Planting / Biodiversity Wadi | 894 m²   |
| 13                                   | Community Plaza & Event Lawn        | 787 m²   |
| 14                                   | Souk Kiosks & Services              | 417 m²   |
| 15                                   | Multipurpose Sports Lawn            | 630 m²   |
| 16                                   | Al Kathib — the dune berm           | 1,464 m² |
| 17                                   | Al Madar — the perimeter loop       | 986 m²   |
| Al Sikkak — shaded alleys & setbacks |                                     | 1,414 m² |

Total 15,000 m²

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.  
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge